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Response to RFI (2025-23641)

Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care

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Cover Letter

Office of the Deputy Secretary and Assistant Secretary for Technology Policy (ASTP) and Office of the National Coordinator for Health Information Technology (ONC) (collectively, ASTP/ONC)
U.S. Department of Health and Human Services
330 C St SW
Floor 7
Washington, DC 20201

To Whom It May Concern:

As technologists, architects, and implementers of systems that contend daily with the complexities of health data, Leap Orbit, LLC, is pleased to respond to ASTP/ONC's Request for Information (RFI) regarding Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care. Our response is predicated on a core belief: that the vision for a digitally integrated healthcare future hinges on getting the foundational data layers right.

In our response, we offer insights into key areas we believe are critical to this mission - from voluntary frameworks to advance market-driven trust, to the need for technical and administrative metadata and its implications for information blocking, to the importance of incentivizing "AI-Ready" data infrastructure. Our experience developing and operating platforms like Convergent, a provider data management (PDM) system that has improved data integrity for our clients at a foundational level, informs these insights, as does our work involving extensive real-world implementation of HL7 Fast Healthcare Interoperability Resources (FHIR) standards, particularly the Da Vinci PDEX Plan Net Implementation Guide (IG). These insights are further informed by our collaborations with federal entities, including the Assistant Secretary for Technology Policy (ASTP), the Health Resources and Services Administration (HRSA) (notably on the UDS+ Modernization initiative), the Substance Abuse and Mental Health Services Administration (SAMHSA), and the Centers for Disease Control and Prevention (CDC).

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We use our work not as a product showcase, but as a case study in how specific technological approaches can address systemic problems and inform sound policies to accelerate AI adoption in clinical settings.

We appreciate the opportunity to contribute to this vital conversation and to share the lessons learned from our efforts to make shared health data accurate, accessible, and useful for all Americans.

Sincerely,

David Finney

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Executive Summary

The Paradox of Connectivity: Why Better Health Data Infrastructure is Non-Negotiable.

The primary bottleneck to clinical AI is no longer the complexity of the algorithms, but the persistent “trust friction” created by fragmented, unverified data pipelines. The core challenge, as Leap Orbit has found in its work, is a misalignment in how we approach foundational data infrastructure. HHS, through this RFI, has wisely focused attention on these underpinnings.

Through hard experience, we have learned that the acceleration of Artificial Intelligence (AI) in clinical care is primarily hindered by a lack of "AI-Ready" foundational data. For AI to be safe, equitable, and effective, it requires a "bedrock" of accurate, accessible, and interoperable data with traceable provenance. Recognizing this need, we recommend that HHS focus its initial efforts on three "Foundational Prerequisites":

- **Market-Driven Trust Frameworks:** Move from prescriptive "black box" oversight toward fostering voluntary, industry-led standards for the integrity and provenance of the data pipelines feeding clinical AI. This approach ensures transparency without the weight of redundant regulatory mandates.
- **Incentivizing Data Integrity**: Structure payment policy to reward "Consent-Aware" and "AI-Ready" infrastructure. By establishing reimbursement for digital consent orchestration and data quality, HHS can incentivize the maintenance of the high-fidelity provider and patient data necessary for autonomous AI to function.
- **Implementation Sandboxes for R&D:** Foster and leverage public-private technical communities to bridge the gap between high-level policy and real-world implementation. These "sandboxes" allow for the rapid iteration of emerging standards, like the Model Context Protocol (MCP) and FHIR, ensuring they are refined by the market before being broadly adopted.

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Introduction: About Leap Orbit

Founded in 2015, Leap Orbit, LLC, is an AI-first U.S.-based healthcare IT firm dedicated to solving intricate health data challenges. We build and implement innovative solutions, including our Convergent Provider Data Management platform and the RxGov prescription drug monitoring (PDMP) suite. Our focus is on leveraging modern, cloud-native architectures—primarily Microsoft Azure—and an Application Programming Interface (API)-priority design philosophy (see developer docs). Increasingly, we see AI not just as a tool, but as a foundational capability that we embed into the core of every product to deliver smarter, more adaptive solutions that improve care and enhance compliance.

Central to our work is a deep commitment to interoperability, exemplified by our leadership in the real-world application of HL7 FHIR standards. This is demonstrated in our contributions to national initiatives like HRSA's UDS+ Modernization and our current and past work with ASTP, SAMHSA, and the CDC. Our Convergent platform, for example, natively supports FHIR R4 and conforms to the Da Vinci PDEX Plan Net IG, serving as a testament to our approach. For clients, this has resulted in successfully standardizing over 98% of provider and facility addresses, a foundational step for accurate directory services.

As a company committed to pushing boundaries and building what’s next, we believe in "glass box" data automation—providing transparency and traceable data provenance—because trust is paramount in health data. Our culture of experimentation and innovation drives our perspective, which is shaped by the technical realities of building and deploying solutions designed to make health data accurate, accessible, and useful. Our commitment to these principles is further underscored by third-party validations such as the Electronic Healthcare Network Accreditation Commission (EHNAC) Privacy & Security Accreditation and DirectTrust Privacy & Security Re-Accreditation for our services.

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Part I: Feedback on Primary Approaches for Federal Action

Regulation: Impact of Current HHS Frameworks on AI Adoption

Current regulations often focus on the clinical end product(s) of an AI tool but neglect the upstream pipeline – i.e., the process by which clinical data is ingested, mapped, and cleaned or transformed before it reaches an AI model. By having limited or no visibility into how the clinical outputs are generated or informed by AI, a vacuum of trust is created among users. The following approaches can markedly improve the upstream data pipeline and therefore mitigate the trust gap that currently impairs AI adoption in clinical care settings:

- **Voluntary Integrity Frameworks to Advance Market-Driven Trust:** In alignment with the deregulatory goals of HTI-5, HHS should pivot away from prescriptive transparency mandates in the Health IT Certification Program and instead focus on encouraging voluntary transparency for the "upstream" pipeline (e.g., data cleansing, identity matching, and provider roster ingestion). Rather than requiring "model cards" or specific source attributes by law, HHS should foster an environment where traceable data provenance and confidence scores are recognized as market-differentiators. In a deregulated environment, the market requires provenance transparency to manage risk. This is achieved through foundational data engineering - specifically, repeatable FHIR ingestion pipelines that "bake" metadata and consent enforcement into the data stream at the point of origin, rather than attempting to fix data quality downstream. Our experience with the Convergent platform and CareLoaDr AI indicates that trust is a prerequisite for automation; users are only willing to adopt AI in day-to-day use when they can see "behind the curtain" of the system to understand specifically how a record was matched or a provider's credentials were verified.

- **Enabling AI-Ready Interoperability through Automated Access:** AI systems require high-fidelity metadata to autonomously "use" clinical data without "special effort". We strongly support the proposed revisions to 45 CFR § 171.102 in HTI-5 to explicitly include automated means, including autonomous AI and agentic systems, within the definitions of "access," "exchange," and "use". This modernization recognizes that AI is now a first-class "actor" in the health data ecosystem.

To fully realize this vision, HHS should clarify, through sub-regulatory guidance under the 21st Century Cures Act information blocking provisions and 45 CFR Part 171, that failure to exchange clinically necessary provenance, integrity, semantic, and consent metadata, when such metadata is reasonably available and necessary for safe automated or clinical use, may constitute a limitation on "use" under the definition of access, exchange, and use (AEU). We recommend that HHS require that the metadata meet the following three criteria to avoid overburdening implementers while enabling innovation:

- Necessary for clinical safety;
- Necessary for interpretability or automation;
- Already representable in existing standards (e.g., FHIR Provenance, Consent, AuditEvent, USCDI elements).

A minimal set of metadata across clinical care AI use cases should include the following categories:

- Provenance metadata (e.g., source, attribution),
- Integrity metadata (e.g., currency, completeness, validation status, encounter context),
- Semantic metadata (e.g., code system, terminology version, transformations, unit normalization, original text for coded elements),
- Sensitivity metadata (e.g., consent status, use restrictions, sensitivity category, redisclosure constraints, revocation timestamp), and
- AI Generation metadata (e.g., AI processing indicator, model identifier, confidence score, human validation status).

Current information blocking concerns often stem from a "black box" approach where data is exchanged without context. Leap Orbit's work with the Convergent platform and the FHIR Da Vinci PDEX Plan Net Implementation Guide (IG) demonstrates that when rich metadata is provided, AI systems can autonomously trace errors back to source files and assign confidence scores. This level of transparency prevents the "misuse and abuse" of information blocking exceptions by ensuring that third-party AI software has the technical context required to be safe and effective.

Finally, we recommend that HHS promote voluntary, standardized metadata headers that includes the minimal set of metadata we describe above for AI-to-AI communication – i.e., a “cover sheet” for every piece of data exchanged between AI agents. An example might be a clinical AI agent designed to predict certain conditions that receives lab results as raw JSON data from disparate sources; having standardized metadata headers that contains provenance, quality scores, and privacy/consent information enables the AI agent to read and digest that data instantly without manual mapping. As a practical application, Leap Orbit recommends that HHS pilot a “Structured Insight Extractor” initiative. This program would utilize Large Language Models (LLMs) to ingest legacy, semi-structured data - such as CCDA clinical summaries or Electronic Lab Reporting (ELR) data - and extract them into high-fidelity FHIR resources. By automating this upstream pipeline, HHS can ensure that clinical AI models are fed standardized data with clear provenance, rather than being forced to infer clinical context from unstructured noise.

- **Standardizing Specialty Data as an Innovation Accelerator**: HHS should prioritize the use of the USCDI+ model (as pioneered in the SAMSHA/ASTP Behavioral Health Information Technology Pilot) to establish voluntary, predictable "data floors" for specialty care. By standardizing these specialty bundles via USCDI+, HHS provides a predictable data floor. This eliminates the need for every AI developer to build custom, site-specific ingestion code, effectively lowering the barrier to entry for innovative clinical tools. The HTI-5 Proposed Rule seeks to reset the Certification Program's scope to prioritize FHIR-

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based APIs that expand the data available for clinical efficiency. USCDI+ strongly aligns with this goal by providing the structured context that Large Language Models (LLMs) and agentic AI systems struggle to infer reliably. Rather than a regulatory checkbox, USCDI+ should be positioned as a market-driven resource that improves AI accuracy and reduces the heavy burden of constant human-in-the-loop validation.

Reimbursement: Levers to Promote AI Adoption

The greatest administrative burden in clinical AI adoption is the "data ingestion" tax – or the cost of cleaning "dirty" data. Payment policy and incentives should be used to lower the burden of this tax and reward the quality of the underlying data ecosystem.

- **Incentivizing "AI-Ready" Data Infrastructure:** CMS should consider including "Data Integrity Scores" as part of Value-Based Care (VBC) benchmarks or Merit-Based Incentive Payment System (MIPS) reporting. For example, organizations that can demonstrate a high level of accuracy in their provider directories and consent frameworks should be eligible for incentive payments. Furthermore, HHS should use payment policy to reward targeted applied machine learning that improves the "ingredients" of the clinical record. Instead of leading with bespoke, high-risk clinical modeling, the private sector should be incentivized to use Data Science capacity to automate normalization, confidence scoring, and identity resolution. Once the underlying data model is stable and identity is resolved with high confidence, the risk profile for downstream clinical AI interventions is significantly lowered. This trusted data foundation has a direct impact on payers, providers, and patients. For payers, it enables compliance with federal mandates and improves operational efficiency; for one national payer, automating CMS 4I reporting through our solutions saved an estimated $240,000 per year and reduced a four-week manual process to under 48 hours. For providers, it reduces the administrative burden of attesting to their information across countless systems. For patients, it means directory tools are reliable, which is crucial for building trust and ensuring access to care. Organizations using autonomous AI

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systems to automate care coordination and reporting should be eligible for higher shared savings or performance bonuses if those systems leverage a standardized, consent-aware infrastructure to improve patient outcomes.

- **Incentivizing AI-Driven "Consent Orchestration"**: AI models, particularly those used for behavioral health integration or sensitive environmental factors affecting health, require data that is often protected by strict privacy laws like 42 CFR Part 2 (substance use) or state-specific mental health statutes. Currently, because there is no standardized, automated way to orchestrate this consent across different systems, this high-value data is often excluded from AI training sets and real-time clinical feeds to avoid compliance risk.

  HHS should consider incentivizing the adoption of FHIR-native consent management (like our Consentric model) by establishing reimbursement codes that reimburse providers for the use of AI tools that demonstrate "Consent-Awareness" - the ability to programmatically respect granular patient restrictions in real-time. By incentivizing the use of automated, standards-based consent solutions, HHS can accelerate AI adoption by providing a scalable, real-time mechanism to ensure that care delivery across networks like TEFCA is always “consent-aware”, thereby reducing legal risk for providers while also allowing patients to maintain control over their sensitive clinical narratives. HHS could operationalize this through a “TEFCA Agentic Orchestrator” initiative. This program would deploy autonomous AI agents across TEFCA to perform real-time, consent-aware data surfacing. By filtering the massive volumes of network-exchanged data to present only the most relevant, authorized information to a clinician, this initiative would demonstrate how AI can move from being a source of noise to a tool used for clinical coordination.

Research & Development: Public Private Partnerships and Pilot Programs

Technical standards and reimbursement incentives alone do not drive adoption; Communities of Practice do. Leap Orbit recommends that HHS invest in “Implementation Science" through national or regional technical communities whose

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mission is to pilot and fine tune FHIR standards, ensuring they are truly "AI ready" before wider implementation.

- **The "Technical Community" Model for R&D:** HHS should consider funding permanent National Technical Communities (modeled after the HRSA UDS+ UTC) to act as iterative implementation sandboxes. These test cooperatives allow federal agencies to test – and revise – emerging FHIR Implementation Guides (IGs) and newer standards like the Model Context Protocol (MCP) with real-world vendors and providers.

  HRSA's UDS+ Unified Test Cooperative is an example of how a group of providers, administrators, and health IT vendors collectively iterated on a new, automated reporting standard leveraging FHIR with high adoption among community health centers (CHCs) in its first year of implementation. Rapidly iterating and refining FHIR IGs through these technical communities will feed the "flywheel" of AI adoption by providing a bedrock of consistent "AI-ready" data necessary to support the autonomous agentic AI systems now recognized under modernized information blocking rules.

- **Public-Private Partnerships for "Specialty AI Bundles":** HHS should consider using pilot programs (similar to the BHIT Pilot Program) to research how specialized datasets, such as behavioral health or sensitive environmental factors affecting health can be harmonized for AI consumption. This "Pilot-to-Standard" approach ensures that federal R&D investments result in deployable, high-impact clinical tools that move beyond "document transport" to the delivery of context-aware data resources that fuel clinical efficiency and reduce provider burnout.

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Part II: Responses to Select Questions

Q2: "What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

Payment Policy: As mentioned above, HHS should establish specific reimbursement pathways that recognize Digital Consent Management as a critical clinical support service – i.e., reimbursing providers that utilize FHIR-native consent orchestration tools. By establishing codes for the real-time orchestration of patient privacy preferences, CMS can provide the financial bridge for providers to move beyond all-or-nothing data exchange and adopt high-value AI interventions that are fully compliant with 42 CFR Part 2. Through this reimbursement, providers are more incentivized to adopt automated, standards-based tools that allow clinical AI to safely scale across networks like TEFCA.

Programmatic Design: Additionally, HHS/CMS should incorporate "Data Readiness and Provenance Scores" into Value-Based Care (VBC) benchmarks and reporting. Currently, organizations lack a financial incentive to invest in technical metadata (e.g., provenance, confidence scores, and semantic tags) that makes clinical records consumable by autonomous systems without "special effort". Prioritizing the use of AI-assisted ingestion (e.g., CareLoaDr AI) would automate the administrative burden of data cleansing while providing "Glass Box" transparency. In an era of deregulated certification (as emphasized in HTI-5) where "model cards" are being removed, the market requires market-driven integrity scores to manage risk and helps to ensure that the appropriate clinical care is being delivered.

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Q5: How can HHS best support private sector activities (e.g., accreditation, certification, industry driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

HHS should position itself as the "Facilitator of Industry-Led Testing & Benchmarking." The most impactful support HHS can provide is to create the "Sandbox Environments" where the private sector can voluntarily test the trustworthiness and interoperability of their AI tools against real-world, standardized datasets. Specifically, HHS can best support the private sector by *funding National Testing Collectives* (modeled after our work with the HRSA UDS+ UTC). These collectives act as a bridge between federal policy and private-sector innovation. Additionally, HHS should support an "AI-Ready" designation - not a mandatory certification, but a voluntary testing badge achieved through these technical communities.

Our work as frontline implementers and deep experience in convening diverse stakeholders proves that industry-driven testing is most successful when it uses a phased, incremental approach to iterate data standards before they are broadly adopted.

Further, as HTI-5 expands the definition of "access" to include autonomous AI and agentic systems, the primary barrier becomes identity and trust. HHS can support the private sector by endorsing open standards that "credential" these agents.

We recommend that HHS promote development and voluntary adoption of a standardized, FHIR-aligned AI Interaction Profile that enables secure, consent-aware retrieval of clinical context and structured return of AI-generated outputs with appropriate provenance and attribution metadata.

Using existing authority under the 21st Century Cures Act certification program and related sub-regulatory guidance, HHS should encourage health IT developers to support

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standardized AI interaction capabilities that prevent proprietary lock-in and ensure auditable, interoperable, and safe AI integration.

HHS should avoid mandating any specific private protocol or requiring disclosure of proprietary algorithms. Instead, it should promote open standards alignment, conformance testing, and reference implementations through cooperative agreements and federal funding mechanisms while allowing emerging standards such as the Model Context Protocol (MCP), which acts as a "universal interface" for AI agents, to mature. MCP allows AI to securely interact with clinical data while maintaining a cryptographically signed audit trail and serves as a standardized “handshake” between AI agents and data sources, such as FHIR - ensuring that context is preserved as information moves between disparate systems. By prioritizing outcomes over methods through standards alignment and testing, HHS can enable the private sector to "credential" their AI tools, ensuring every action is permissioned, auditable, and traceable, without foreclosing industry innovation and refinement.

Q8: “Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.”

The scalability trap must be solved for AI to move from a "bespoke pilot" to a scalable clinical utility. Currently, every new AI deployment requires custom, site-specific data mapping that creates a significant training debt. Enhanced interoperability fuels AI by transforming fragmented information into “clean” data, i.e., data that is normalized, attributed, and consent-cleared at the point of ingestion.

Enhanced interoperability widens market opportunities by eliminating the “bespoke training debt” typical of current AI deployments. By prioritizing repeatable FHIR ingestion pipelines, developers can ensure that provenance and consent are handled programmatically. This allows AI teams to focus on targeted applied machine learning for identity resolution and data stability, rather than spending much of their R&D

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budget on site-specific data cleansing. This approach, moving from custom code to repeatable pipelines, is the key to making clinical AI deflationary and scalable.

When tools like CareLoaDr AI can use "Glass Box" machine learning to automate the mapping of complex provider rosters into standardized FHIR formats, it reduces the cost of entry for all AI developers. To catalyze this shift, we propose a "National Golden Provider Record" initiative. This multi-payer pilot would use AI to harmonize disparate administrative rosters and credentialing feeds into a single, trusted source of truth. By resolving provider identity at the point of ingestion, HHS would solve the scalability trap for the market, providing AI developers with a predictable foundation of accurate provider data and eliminating the need for redundant, site-specific identity resolution.

Enhanced interoperability also fuels research that supports whole-person care, as "plus" categories of interoperability, such as behavioral health or sensitive environmental factors affecting health are often hampered by data gaps. We recommend *accelerating the domain-specific extensions of USCDI+*. Our experience managing the SAMHSA/ASTP BHIT Pilot shows that these "specialty bundles" are the essential ingredients for research-grade AI.

Additionally, to fuel large-scale research, HHS should *prioritize Bulk FHIR for population-level datasets*, ensuring AI models can be trained on representative, cross-system data without manual intervention.

Conclusion

The task of accelerating AI adoption as part of clinical care is a task of getting the fundamentals right. While emerging technologies capture the imagination, Leap Orbit's experience building foundational data systems like Convergent reinforces the lesson that sustainable innovation rests upon a bedrock of accurate, accessible, and interoperable core information. The challenges in achieving this for data as complex and dynamic as all providers are significant, but as our own experience shows, they are not insurmountable.

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As Leap Orbit’s experience with ASTP, HRSA, and the CDC demonstrates, the path to a high-performing, AI-enabled health system begins with a relentless focus on the foundational data architecture. By addressing the "prerequisites" of identity, consent, and data integrity, HHS can ensure that AI becomes a trusted partner in clinical care rather than a source of administrative complexity. The questions posed in this RFI are central to that endeavor. As a company committed to tackling these complex data challenges, we can collectively and incrementally shepherd approaches that support AI adoption in clinical care (and/or adjacent) settings. We welcome continued engagement on how to best achieve these shared, critical objectives.